

1. Linux-Introduction

What is Linux?

Linux is a free and open-source operating system based on the Unix architecture. It is widely used in servers, cloud computing, embedded systems, and enterprise environments due to its security, stability, and flexibility.

Features

- Open Source
- Multiuser Operating System
- Multitasking
- Secure
- Portable
- Stable
- Highly Customizable

Linux Distributions

- Ubuntu
- CentOS
- Red Hat Enterprise Linux
- Debian
- Fedora

Applications

- Cloud Computing
- DevOps
- Web Servers
- Database Servers
- AI & Machine Learning

2. Linux-Filesystem

Important Directories

- `/` Root Directory
- `/home` User Files
- `/bin` Essential Commands
- `/etc` Configuration Files
- `/var` Log Files
- `/tmp` Temporary Files
- `/usr` Installed Programs

File Types

- Regular Files
- Directories
- Symbolic Links
- Devices

Benefits

- Organized Storage
- Better Security
- Easy Navigation

3. Linux-Commands

Navigation

```
pwd  
ls  
cd
```

File Operations

```
touch  
cp  
mv  
rm  
mkdir  
rmdir
```

Viewing Files

```
cat  
less  
head  
tail
```

Search

```
find  
grep
```

Permissions

```
chmod  
chown
```

4. Linux-Users

User Management

User Commands

```
useradd  
passwd  
usermod  
userdel
```

Group Commands

```
groupadd  
groupdel  
groups
```

User Types

- Root User
- Regular User
- System User

Importance

- Security
- Access Control
- Resource Management

5. Linux-Process

Process Management

Commands

```
ps  
top  
htop  
kill  
killall  
nice
```

Process States

- Running
- Waiting
- Sleeping
- Zombie

Purpose

Monitor and control running programs efficiently.